

Rasmika Billa

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EDUCATION

NYU - Courant, New York

Sep 2024 – May 2026

Master of Science in Computer Science, AI Concentration, GPA: 3.7/4.0

Vellore Institute of Technology, India

Aug 2018 – Jul 2022

Bachelor of Technology in Computer Science, specialization in Data Analytics, GPA: 3.9/4.0

SKILLS

Languages: Python, R, SQL, PostgreSQL, Pyspark, C/C++, JavaScript, Typescript, React, Next.js, Node.js

AI/ML: PyTorch, Tensorflow, LangChain, LangGraph, NLP, Predictive Modelling, XGBoost, scikit-learn, LLM, RAG

Developer Tools: Git, Docker, AWS, Google Cloud, Azure, Databricks, Linux, GitHub Actions, Kubernetes, CI/CD

Engineering Tools: ETL/ELT, Power BI, Tableau, Snowflake, Databricks, Rest API, FastAPI, Cursor, Claude Code

Frameworks/Databases: Flask, Streamlit, MongoDB, Hadoop, Hive, Pinecone, BigQuery, MLOps, A/B Tests

EXPERIENCE

AI Engineer Intern - GEP (*Consulting and Strategy Team*)

June 2026 – Present

- Designed a multi-agent, LLM-driven should-cost modeling pipeline with tool calling, prompt engineering, and human-in-the-loop approvals, mapping 100+ spend categories to 40+ market indices for Monogram Foods.
- Built a semantic knowledge graph (Oxigraph, Apache AGE) to ground LLM agents, reducing retrieval errors by 26%.

AI Engineer Intern - Interplay (*Ascend FO Team*)

May 2025 – Aug 2025

- Deployed LLM-powered Agentic AI workflows to automate investment reporting across 14+ client portfolios, reducing turnaround time by 80%; translated portfolio manager requirements into production-grade data pipelines.
- Implemented Differential Evolution-based asset allocation optimization model to maximize Net-CAGR up to 8%.

Data Scientist - Royal Dutch Shell (*Asset Management Team*)

Aug 2022 – Aug 2024

- Engineered a Retrieval Augmented Generation (RAG) system using LangChain, custom API, and SQL in collaboration with safety and operations teams, achieving 86% model accuracy in report quality evaluation.
- Developed a chatbot using GenAI (GPT API) tailored to each terminal's safety data for efficient recommendation.
- Architected data pipelines in Azure Data Factory for catalyst monitoring, contributing to a 4% revenue growth.
- Transformed unstructured data using OCR, reducing human error and saving 3,400+ work-hours at the plant site.

Data Analyst - Tenderscuts (*Business Intelligence Team*)

Aug 2021 – Jul 2022

- Predicted weekly order volumes with Decision Trees for each store, leading to a 3% reduction in package usage.
- Conducted RFM analysis and built recommendation system that improved customer targeting effectiveness by 93%.

PROJECTS

Credit Guard Trained and calibrated a financial risk model to predict 12-month loan default probability for a large Italian bank, achieving AUC 0.797 with XGBoost and improved robustness through influential observation analysis.

DeepDive Constructed an agentic research assistant using LangGraph, orchestrating a 3-node graph (search, summarize, report) with a local LLM and web search; developed a custom evaluation script scoring relevance, coverage, and quality.

PLAS Quantified platelet activity from EHR clinical data using Quantile Regression, benchmarking against XGBoost via 5-fold cross-validation and reaching 78% prediction accuracy, outperforming baseline model by 6%.

RESEARCH/PATENT

AI For Factor Investing: A Comprehensive Survey (*Springer Nature, ICI AI*)

Jan 2026

- Surveyed ML, RL, and GenAI methods for factor discovery, portfolio construction, trading, and risk control.
- Identified limitations of AI-based factor models and proposed strategies to address overfitting and interpretability.

Scheduling for Healthcare Centre for COVID-19 (*InHeNce - ID: 1570726833*)

Sep 2021

- Analyzed initial covid flow with NetworkX and produced a Bi-LSTM model with an MSE score of 7.2174e-04.
- Created a tool to optimize medical staff allocation using Genetic algorithm based on next week's predicted cases.

Py - Digital Writing Pad (*Patent App no. 202141026740*)

Jun 2021

- Devised a motion-based tool with Transformer Architecture enabling users to capture ideas on-screen with any stylus.
- Achieved 72% reduction in costs by redesigning a pad utilizing **Deep Average Network** for hand motion detection.